

Laboratory Report 1: Microscopy and Measurement

Due: Wednesday, September 23 at 11:59 PM

Value: 100 points

Format: Individual scientific report

Purpose

Determine whether two plant tissues differ in mean cell length while demonstrating correct calibration, measurement, and uncertainty reporting.

Your task

- Convert ocular divisions to micrometers using 10 ocular divisions = 25 micrometers.
- Calculate cell length for all 12 cells, then report mean, range, and sample size for each tissue.
- Create a dot plot or box plot showing every measurement, not only the means.
- Write a 1,200-1,500 word report with question, methods summary, results, interpretation, limitation, and follow-up.
- Evaluate whether field selection or cell orientation could bias the observed difference.

Provided evidence or data

Cell	Onion epidermis (ocular div.)	Elodea leaf (ocular div.)
1	31	18
2	35	21
3	29	20
4	38	19
5	33	23
6	36	22
7	32	20
8	34	24
9	37	21
10	30	19
11	35	22
12	33	20

Submit

- Scientific report PDF with one figure and one concise results table.
- Appendix containing converted cell-level data and sample calculations.

Important note

These are simulated instructional measurements. Do not report inferential significance; focus on estimation, variation, and measurement limitations.

Grading criteria

Criterion	Pts	Full-credit evidence
Question and methods	15	Frames the investigation and describes the relevant procedure accurately.
Results	25	Calculations, figures, captions, and statistical summaries are correct and transparent.
Interpretation	30	Explains results using an appropriate cellular mechanism and addresses uncertainty.
Evaluation	20	Identifies limitations, controls, and a justified follow-up experiment.
Scientific form	10	Clear structure, accurate units, citations, and reproducible data presentation.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.